Courier Delivery Analysis

Operational and Financial Analysis of Bicycle Deliveries

MAGDALENA BAŁABAŃSKA| GITHUB: MAGDUS-DATA-SCIENCE

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Executive summary

This project analyzed operational and financial aspects of bicycle courier deliveries in urban food delivery services using Python, R, statistical analysis, and machine learning techniques.

The dataset consisted of 451 bicycle deliveries completed in Cracow, Poland over a 44-day observation period between December 2025 and April 2026. Operational, financial, temporal, and weather-related variables were manually collected and integrated into a structured analytical dataset.

The exploratory analysis revealed strong temporal and environmental patterns in courier operations. Evening hours generated the highest earnings, while adverse weather conditions increased operational difficulty and delivery duration. Weekend deliveries produced higher average earnings and tips compared to weekdays.

Statistical analysis confirmed that some weather-related variables significantly influenced delivery duration. Linear regression results indicated that higher temperatures were associated with shorter delivery times, while rainfall showed an unexpected negative relationship with delivery duration. However, the model explained only a limited portion of delivery time variability, suggesting that additional operational factors such as traffic conditions, route distance, and restaurant preparation time play an important role.

Machine learning models were developed to predict delivery duration. Random Forest outperformed the baseline Linear Regression model, reducing the average prediction error to approximately 7 minutes. Despite moderate predictive performance, the relatively low R^2 score indicated that delivery operations are influenced by substantial real-world variability not captured in the available dataset.

From a business perspective, the analysis suggests that:

- evening hours represent the most operationally profitable period,
- weather conditions significantly affect courier efficiency,
- dynamic staffing and incentive strategies could improve platform performance during difficult operational periods.

Future improvements could include integrating GPS route data, real-time traffic information, restaurant waiting times, and advanced boosting algorithms to improve predictive accuracy and operational insights.

1. Introduction

The objective of this project was to analyze factors influencing courier earnings and delivery efficiency in urban food delivery operations.

2. Dataset Overview

Data frame description

The dataset contains 451 bicycle courier deliveries completed over 44 working days between December 2025 and April 2026 in Cracow, Poland. After each shift, operational data was manually recorded and later converted from Excel format to CSV. Data came from:

- App "Glovo Rider": Order time, earnings and type of delivery data
- http://meteo.fis.agh.edu.pl/archivalCharts : Weather data
- Personal notes: Target districts data

What are the variables in the raw data:

- ORDER TIME
  - **time start**- time of accepting the order
  - **time end**- time of delivery confirmation
- WEATHER
  - **temp**- temperature in Celsius degrees
  - **fall**- in mm/h
  - **wind**- mean in m/s
- EARNINGS- in zloty
  - **basic**- salary calculated with distance to partner, then to customer. Shown multiplicated by a factor depending on traffic. Factor rises, if weather's become bad (falls).
  - **factor**- depends on traffic and weather.
  - **tip**- extra money from the customer.
  - **total**- tip added to the basic earning.

Earnings do not contain prizes from challenges and distance adjustments.

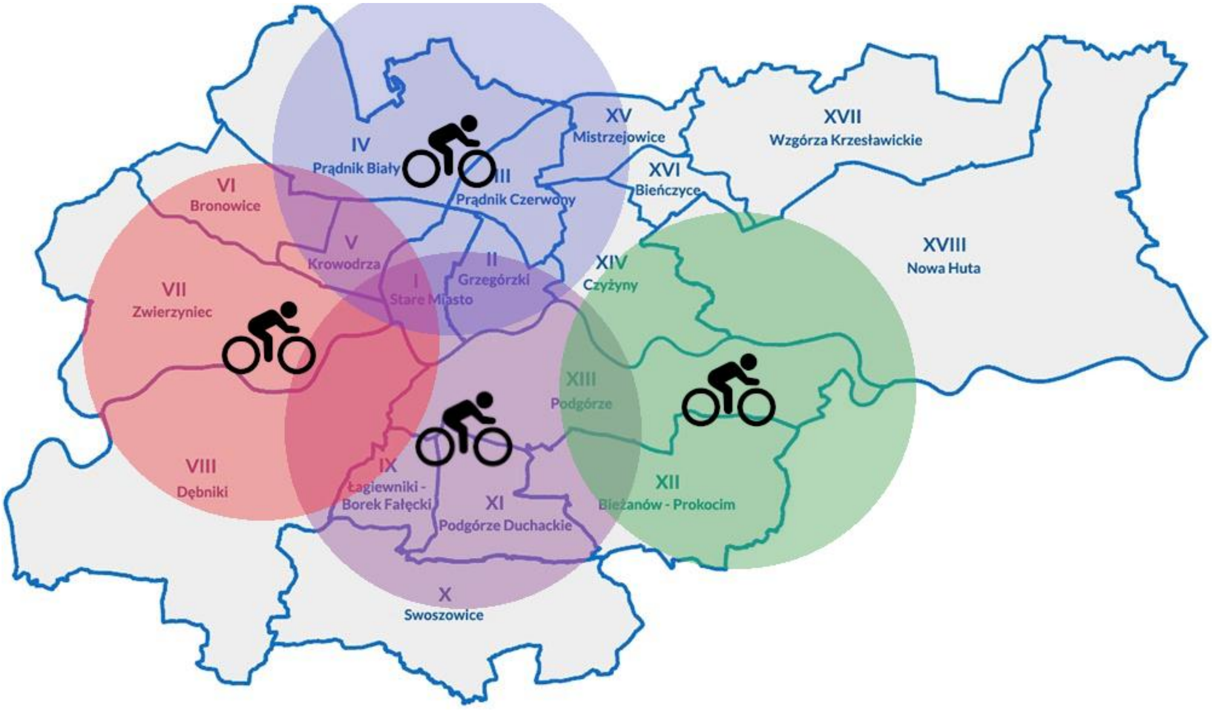
TYPE OF DELIVERY (alphabetical)- what do customers order?

animal goods	goods from animal store (including food)
bakery	bread, buns and some pastry
breakfast	dishes usually eaten in the morning
burgers	from burger- oriented diner
coffee	from coffee store
chinese	dishes originated from China
deli	ready-to-eat or minimally heat-treated (e.g. heating) dishes, snacks and semi-finished products
fast food	various dishes including fries, burgers, nuggets, 'zapiekanka'
flowers	flowers and gift shop goods
french	dishes originated from France
groceries	various goods bought in supermarket (mostly food)
indian	dishes originated from India
italian	dishes originated from Italy (excl. pizza)
japanese	dishes originated from Japan
kebab	from kebab- oriented diner
korean	dishes originated from Korea
liquors	alcohol beverages
mexican	dishes originated from Mexico
pharma	OTC medications, supplements, cosmetics and minor medical supplies
pick up	a transport service for small items between addresses
pizza	from pizza- oriented diner
polish	dishes originated from Poland
portugese	dishes originated from Portugal
salads	from salad- oriented diner
spanish	dishes originated from Spain
tea	from tea store, mostly bubble tea
turkish	dishes originated from Turkiye (excl. kebab)
ukrainian	dishes originated from Ukraine
vietnamese	dishes originated from Vietnam

TARGET DISTRICT- where do the customers live?

District	Symbol	District	Symbol
Stare Miasto	I	Swoszowice	X
Grzegórzki	II	Podgórze Duchackie	XI
Prądnik Czerwony	III	Bieżanów- Prokocim	XII
Prądnik Biały	IV	Podgórze	XIII
Krowodrza	V	Czyżyny	XIV
Bronowice	VI	Mistrzejowice	XV
Zwierzyniec	VII	Bieńczyce	XVI
Dębniki	VIII	Wzgórza Krzesławickie	XVII
Łagiewniki- Borek Fałęcki	IX	Nowa Huta	XVIII

Bicycle couriers have distance limit in their orders. Operating in Prądnik Czerwony (the northern part of city), increase the probability of receiving deliveries to nearby districts. Like on the map below. The ranges there are approximate.



In the feature engineering process, the new variables have added:

- **Weekday**- from date
- **Is weekend**- a Boolean variable, TRUE if weekday was Saturday or Sunday
- **Hour**- from time start, used for grouping orders
- **Duration minutes**- from time start and end
- **Bad weather flag**- another Boolean variable, TRUE, if at **least one** of the given conditions was true:
  - Fall > 0,
  - Temp < 0 **or**
  - Wind > 20

Screenshot dataframe

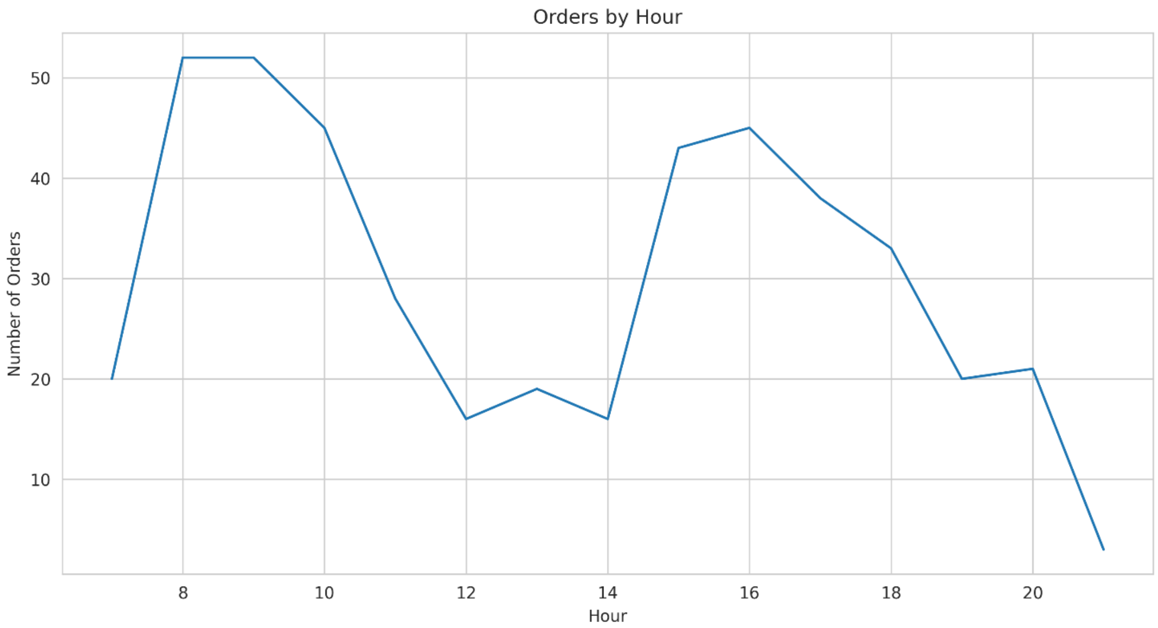
index	date	weekday	hour	time_start	time_end	duration_minutes	temp	fall	wind	bad_weather_flag	basic	factor	tip	total	type	target	tar_sym	is_weekend
0	2025-12-02	Tuesday	9	09:04:00	09:46:00	42	1.0	0.0	1.0	false	18.05	1.69	0.0	18.05	breakfast	Stare Miasto	I	false
1	2025-12-02	Tuesday	9	09:46:00	10:19:00	33	1.0	0.0	1.8	false	12.19	1.69	0.0	12.19	breakfast	Podgórze	XIII	false
2	2025-12-02	Tuesday	10	10:20:00	11:04:00	44	1.5	0.0	0.9	false	14.34	1.52	0.0	14.34	coffee	Krowodrza	V	false
3	2025-12-02	Tuesday	11	11:50:00	12:17:00	27	2.0	0.0	1.7	false	11.55	1.67	2.25	13.8	vietnamese	Prądnik Biały	IV	false
4	2025-12-02	Tuesday	12	12:17:00	13:02:00	45	2.5	0.0	1.2	false	17.53	1.97	0.0	17.53	groceries	Bronowice	VI	false
5	2025-12-02	Tuesday	13	13:02:00	13:43:00	41	2.5	0.0	1.5	false	17.63	2.2	0.0	17.63	liquors	Zwierzyniec	VII	false
6	2025-12-02	Tuesday	17	17:06:00	17:52:00	46	2.5	0.0	0.6	false	14.69	2.2	0.0	14.69	pizza	Stare Miasto	I	false
7	2025-12-02	Tuesday	17	17:52:00	18:22:00	30	2.0	0.0	0.8	false	18.8	2.2	0.0	18.8	fast food	Grzegórzki	II	false
8	2025-12-02	Tuesday	18	18:43:00	19:39:00	56	1.6	0.0	0.9	false	22.37	2.2	0.0	22.37	burgers	Prądnik Czerwony	III	false
9	2025-12-02	Tuesday	18	18:56:00	19:45:00	49	1.6	0.0	0.8	false	11.91	2.2	3.98	15.89	tea	Prądnik Czerwony	III	false
10	2025-12-04	Thursday	9	09:01:00	09:31:00	30	0.0	0.0	1.2	false	9.5	1.41	0.0	9.5	groceries	Prądnik Czerwony	III	false
11	2025-12-04	Thursday	9	09:31:00	09:51:00	20	1.0	0.0	0.7	false	8.84	1.41	0.0	8.84	groceries	Prądnik Czerwony	III	false
12	2025-12-04	Thursday	9	09:51:00	10:09:00	18	1.8	0.0	0.8	false	10.27	1.41	0.0	10.27	fast food	Prądnik Czerwony	III	false
13	2025-12-04	Thursday	10	10:55:00	11:18:00	23	2.9	0.0	1.3	false	10.14	1.41	0.0	10.14	flowers	Prądnik Czerwony	III	false
14	2025-12-04	Thursday	11	11:27:00	11:57:00	30	4.0	0.0	1.4	false	11.96	1.56	0.0	11.96	italian	Prądnik Biały	IV	false
15	2025-12-04	Thursday	11	11:57:00	12:18:00	21	4.8	0.0	0.3	false	8.44	1.56	0.0	8.44	groceries	Prądnik Czerwony	III	false
16	2025-12-04	Thursday	12	12:19:00	13:01:00	42	4.9	0.0	0.4	false	21.06	1.88	3.79	24.85	kebab	Krowodrza	V	false
17	2025-12-04	Thursday	13	13:07:00	13:56:00	49	5.2	0.0	0.9	false	20.16	2.03	0.0	20.16	pizza	Stare Miasto	I	false
18	2025-12-04	Thursday	13	13:56:00	14:27:00	31	3.8	0.0	0.8	false	19.13	2.03	0.0	19.13	fast food	Krowodrza	V	false
19	2025-12-04	Thursday	17	17:00:00	17:30:00	30	1.6	0.0	0.5	false	17.21	2.14	0.0	17.21	pizza	Stare Miasto	I	false
20	2025-12-04	Thursday	17	17:30:00	18:06:00	36	1.3	0.0	0.3	false	14.37	2.14	0.0	14.37	deli	Krowodrza	V	false
21	2025-12-04	Thursday	18	18:06:00	18:31:00	25	1.5	0.0	0.4	false	14.03	2.18	0.0	14.03	fast food	Zwierzyniec	VII	false

3. Exploratory Data Analysis

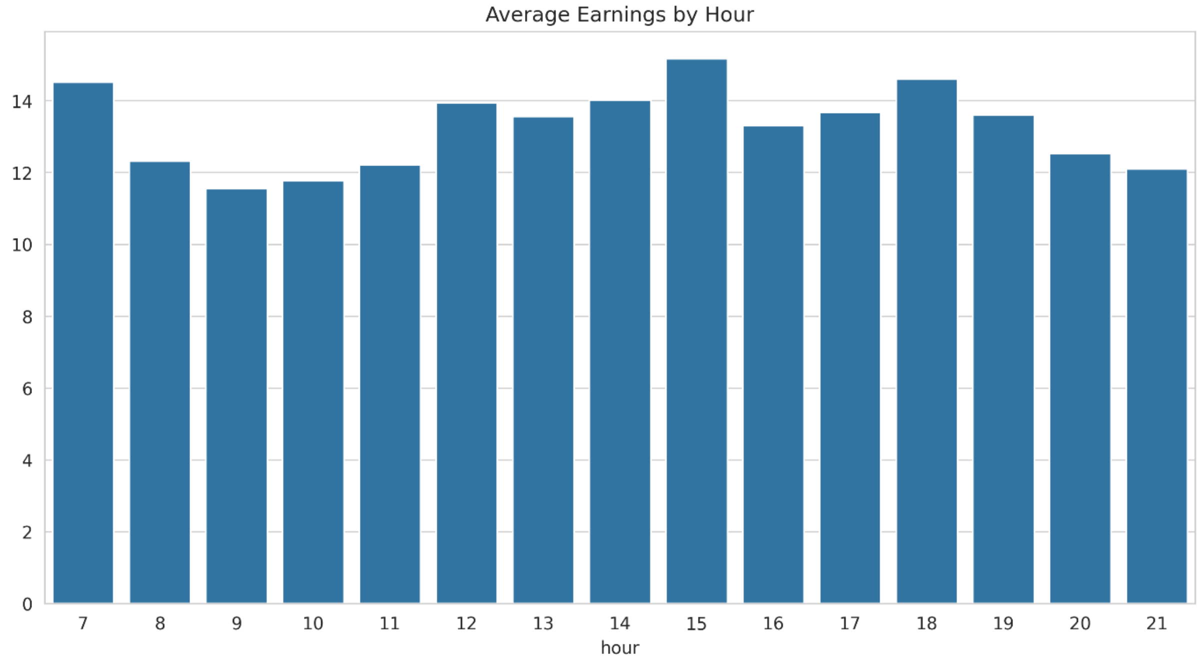
KPI’s

KPI	Value
Orders	451
Avg duration	29 min
Avg earnings (per order)	13 PLN
Avg tip	0.62 PLN
Observation period	44 days

Distributions

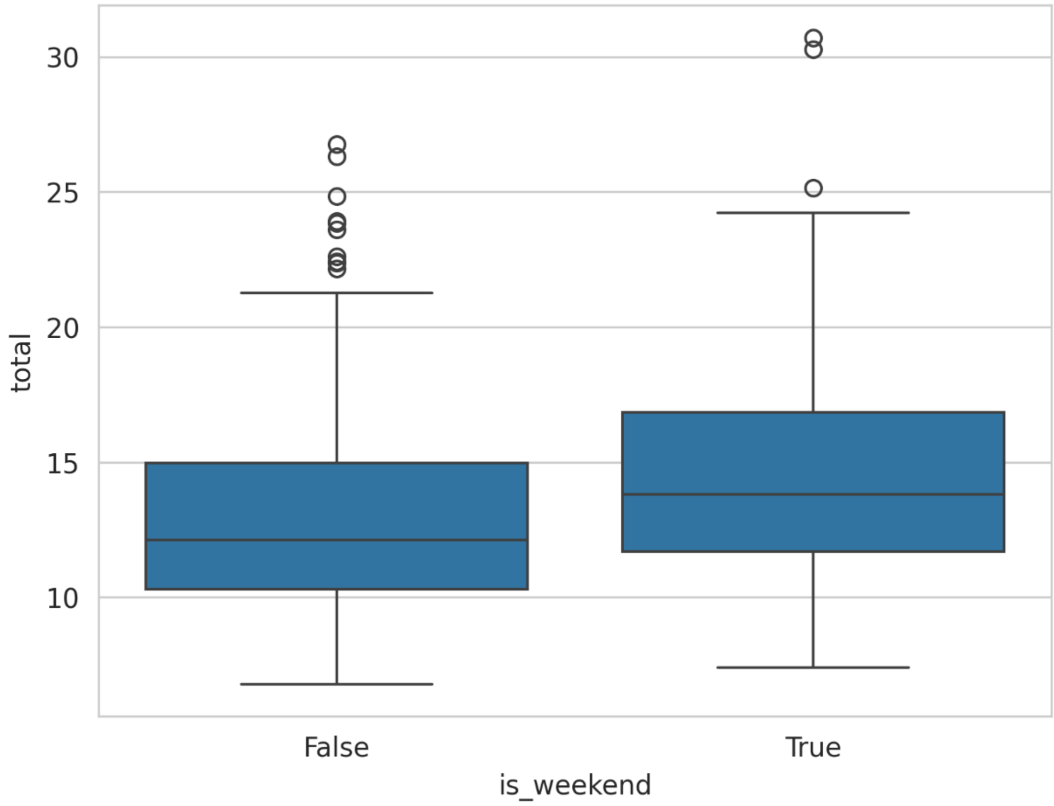


The delivery app operates from 7 AM till 4 AM next day. Courier shifts were most frequently available during early morning and evening hours. Between 12 and 2 PM I usually had a lunch break after 3-4 hours of intensive ride. Afternoon and evening sessions typically offered higher multiplier factors and therefore higher earnings.



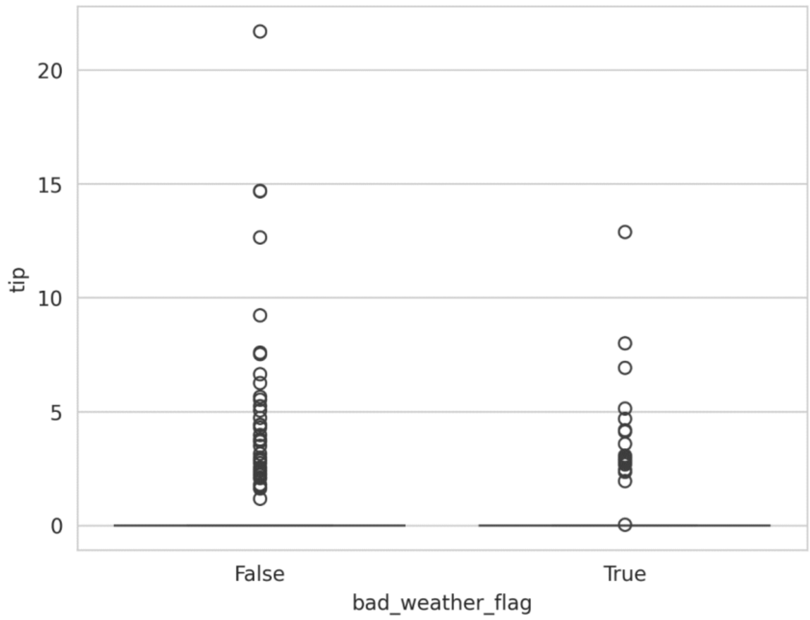
Although multiplier factors were relatively low at 7 AM, reduced courier availability increased the likelihood of receiving long-distance deliveries. Earnings increased gradually throughout the afternoon and peaked during evening hours.

Weekend vs working weekdays

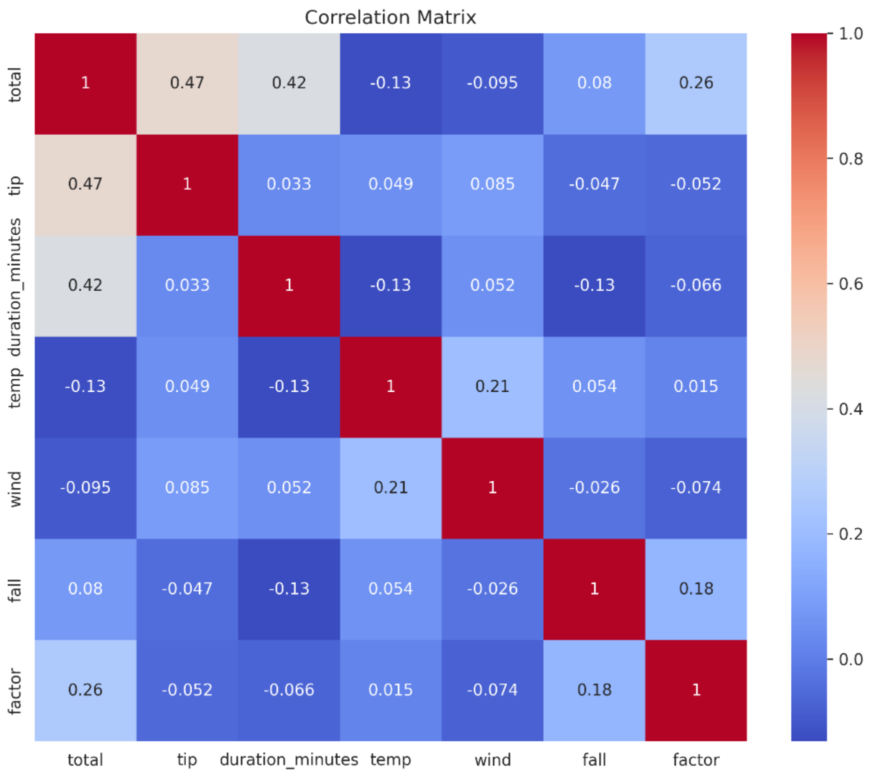


Weekend is more profitable than working weekdays.

Weather analysis

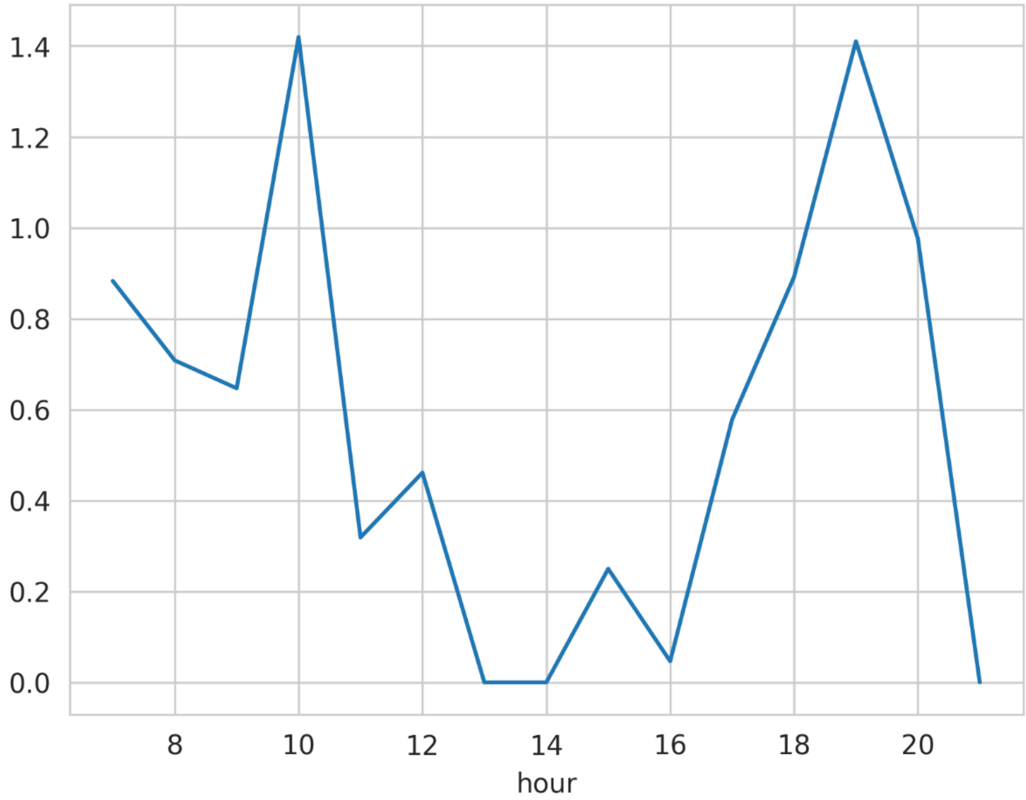


A higher average tip value was observed during favorable weather conditions. Further behavioral analysis would be required to explain this pattern.



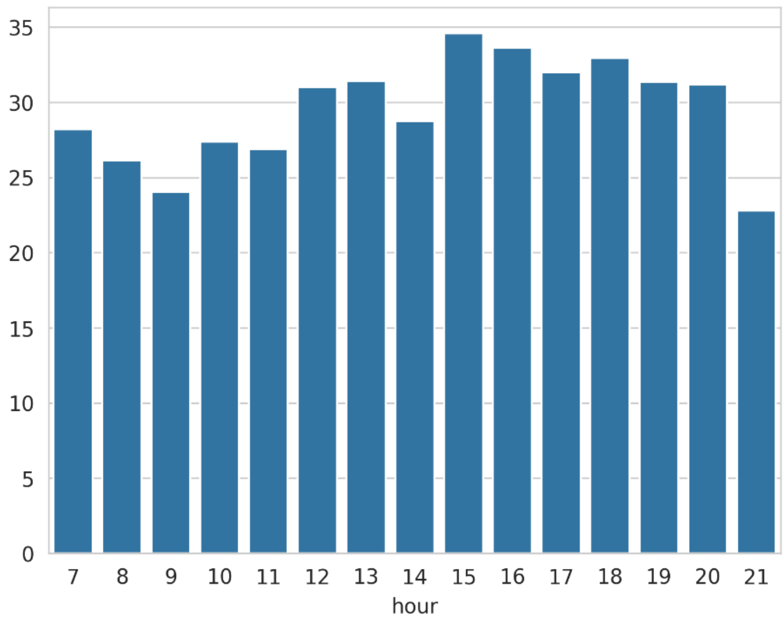
The main driver of change in this dataset is the total variable, which exhibits the strongest (moderate) correlations with tip and duration. Weather factors have negligible or very weak influence on the financial and time variables.

Tips analysis



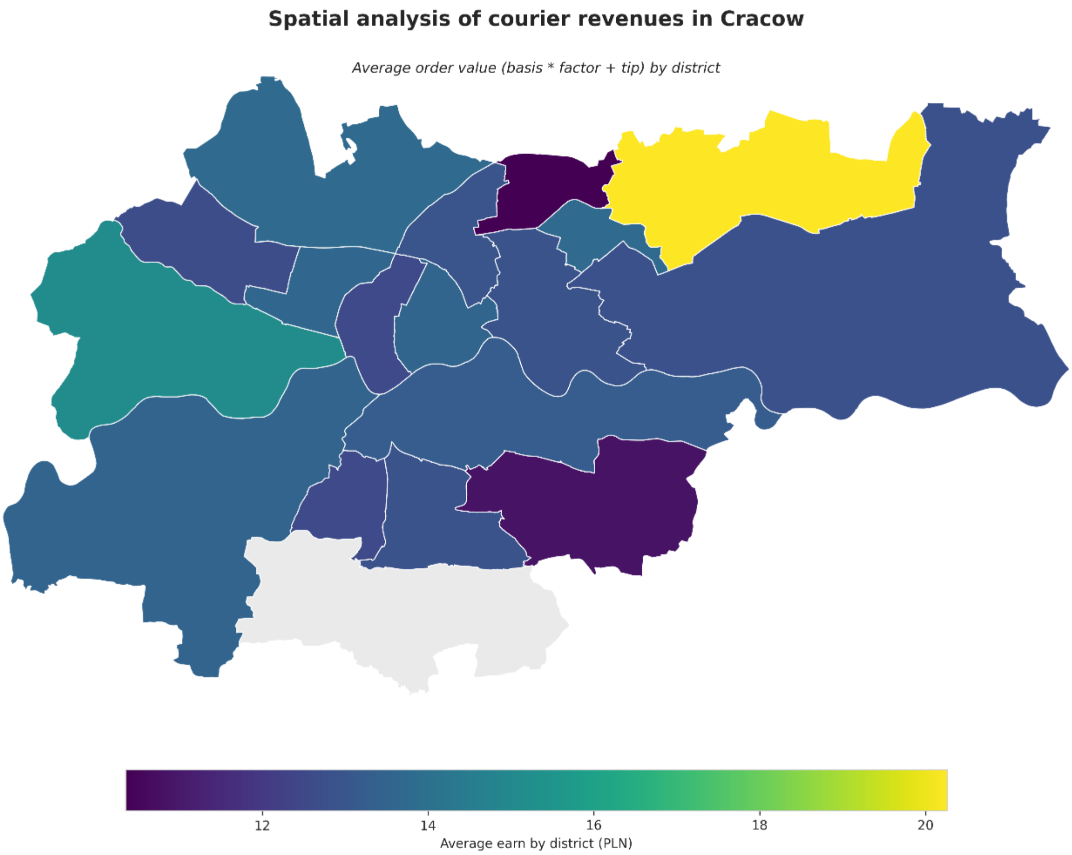
Most tipped are the evening hours.

Earnings per hour

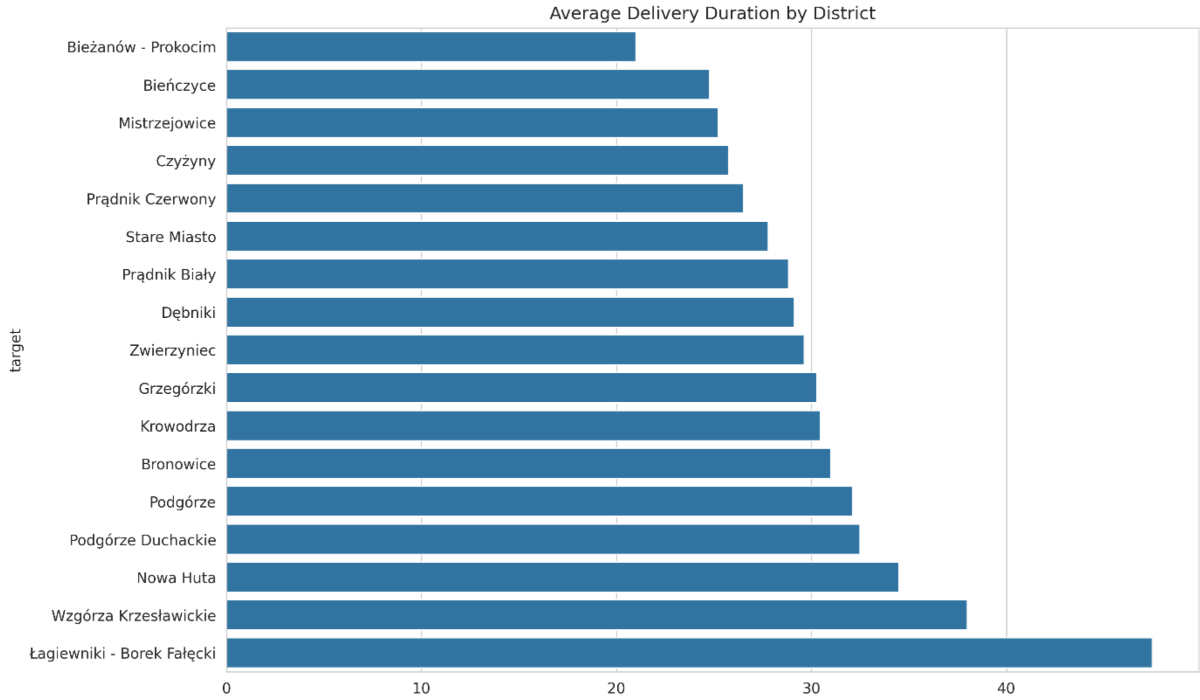


Late afternoon/early evening (3-8 PM) hours are most profitable. The decent earnings are also at noon (12-1 PM).

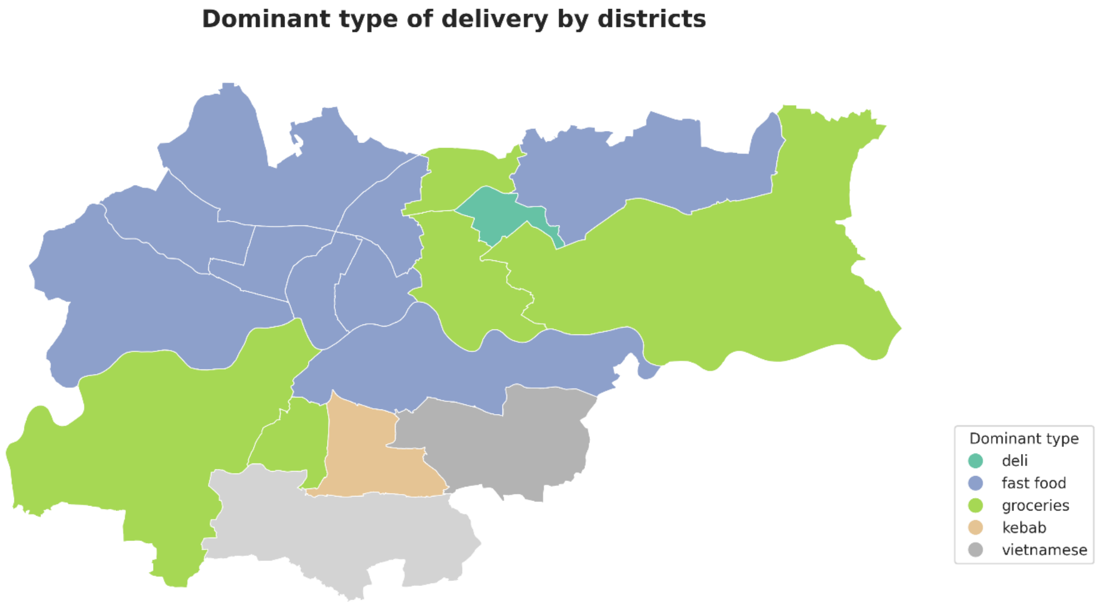
Geographic analysis



The only link between earnings and localization is distance between courier and customer. As above- operating in one district increase the probability of receiving deliveries to nearby districts.



I was in Podgórze district when the order to Bieżanów- Prokocim appeared. This likely contributed to the relatively short average delivery duration observed in that district.



The big fast-food chains introduce many coupons to use in delivery apps. So, in near half of the city dominant type of delivery is fast food dishes. In the distant parts of the city groceries are ordered the most.

4. Statistical Analysis

Hypothesis I:

Earnings in weekends are bigger than in the other weekdays

is_weekend	avg_total	avg_duration
<lgl>	<dbl>	<dbl>
FALSE	12.89598	28.57474
TRUE	14.71889	28.31746

T- test

```
t = -2.8439, df = 74.177, p-value = 0.005756
```

Weekend deliveries generated significantly different earnings compared to weekdays ($p < 0.05$).

Hypothesis II:

Delivery duration is influenced by weather conditions.

Linear regression

```
Residual standard error: 9.483 on 445 degrees of freedom
Multiple R-squared:  0.03962,
Adjusted R-squared:  0.02883
F-statistic: 3.671 on 5 and 445 DF,
p-value: 0.002886
```

Linear regression analysis indicated that temperature and rainfall were statistically significant predictors of delivery duration. Higher temperatures were associated with shorter delivery times, while rainfall unexpectedly showed a negative relationship with duration. Other variables, including wind speed, weekend status, and delivery multiplier, were not statistically significant predictors.

Despite statistical significance, the model explained only a small portion of delivery duration variability ($R^2 \approx 0.04$), suggesting that additional operational factors such as traffic conditions and route characteristics likely play a major role.

Hypothesis III:

Duration of delivery is influenced by a part of a day

ANOVA

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
factor(hour)	14	1257	89.81	0.969	0.0484
Residuals	436	40413	92.69		

Despite visible variation across hours, ANOVA results indicated no statistically significant differences in average delivery duration.

It appears to depend more on operational and environmental factors than on time-of-day alone.

5. Machine Learning

The objective of the machine learning section was to identify factors influencing delivery duration and evaluate predictive model performance. Features used here were:

- Time
- Temperature
- Wind speed
- Fall intensity
- Factor
- Day of the week
- District
- Type of delivery

The linear regression model and random forest model have been deployed in this case. The figures of evaluation are shown in a table below.

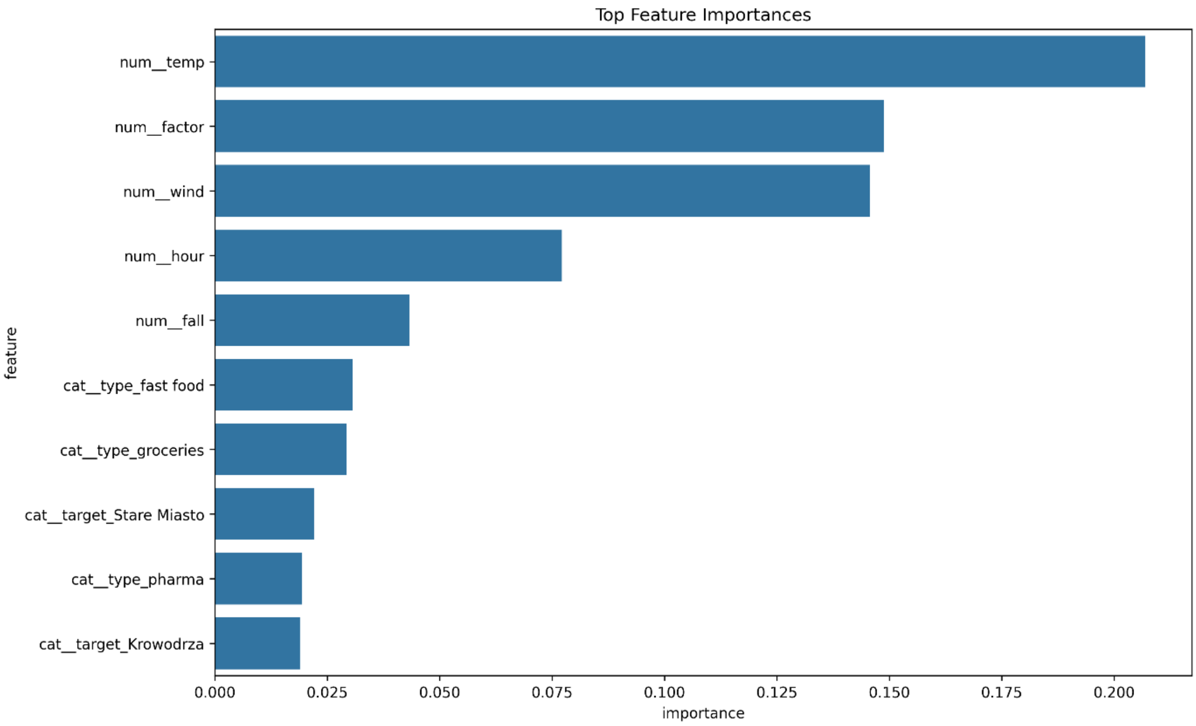
Evaluation

Model	MAE	RMSE	R^2
Linear Regression	7.6	9.9	-0.08
Random Forest	7.0	9.1	0.08

Conclusion

The Random Forest model outperformed the baseline linear regression model, reducing average prediction error to approximately 7 minutes.
Cross-validation confirmed stable model performance across different train-test splits.
However, the relatively low R^2 score suggests that delivery duration depends on additional external factors not included in the dataset.

Feature importance plot



The most influential features in predicting delivery duration were temperature, wind speed, and multiplier factor.

6. Conclusions

This project analyzed operational and financial aspects of bicycle courier deliveries using Python, R and machine learning techniques.

The analysis revealed that delivery efficiency depends strongly on time-of-day patterns and weather conditions. Evening hours generated the highest earnings, while adverse weather increased delivery duration and operational difficulty.

Machine learning models achieved moderate predictive performance, with Random Forest outperforming the baseline regression model. The relatively low R^2 score suggests that delivery duration is influenced by additional external variables not included in the dataset, such as traffic conditions or restaurant preparation time.

From a business perspective, the results suggest that dynamic staffing strategies and weather-based incentives could improve courier efficiency and platform performance.

Future improvements could include route-level geographic data and advanced boosting algorithms to enhance predictive accuracy.

7. Future Work

Some improvements can be employed in the next versions of the project:

- Traffic data,
- Route distance,
- Advanced boosting algorithms,
- Measured restaurant time